VWO Login Page - Product Requirement Document (PRD)

Page: https://app.vwo.com/#/login

Product: VWO (Visual Website Optimizer), by Wingify

Version: 0.1 draft

Written for: QA engineers and product owners deriving test cases from the login feature.

> **Evidence note.** The page is a JavaScript single-page app. An automated fetch returned only the app shell (title "Wingify - Application", loader, unfilled `{{helpTooltipTitle}}` placeholder) and no login form. Everything below is **inferred** from typical VWO sign-in behaviour and standard login practice, not observed on the live page. Each requirement is tagged `[Inferred]` or `[Confirm]`. Verify against the live page and the product team before sign-off.

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1. Purpose

Let registered VWO users authenticate and reach their account dashboard, where they run A/B tests, personalisation, heatmaps, session recordings and other optimisation features.

2. Goals

- G1. Registered user signs in with minimal friction.
- G2. Prevent unauthorised access to account and experiment data.
- G3. Give clear recovery paths (forgot password, SSO, support).
- G4. Support enterprise login (SSO, 2FA).

3. Non-goals

- Account registration and trial sign-up flow (separate PRD, linked from login page only).
- Billing, user and role management after login.

4. Users and personas

Persona	Need
Marketer / CRO specialist	Fast daily login, stay signed in
Developer / QA	Log in on test accounts, repeat logins across environments
Enterprise admin	Log in through company SSO, enforce 2FA
New or locked-out user	Reset password, get help

5. Scope: page elements

ID	Element	Notes	Tag
E1	VWO / Wingify logo	Branding header	[Inferred]
E2	Email field	Account email as username	[Inferred]
E3	Password field	Masked, show/hide toggle	[Inferred]
E4	Remember me checkbox	Keeps session longer	[Confirm]
E5	Sign in button	Submits credentials	[Inferred]
E6	Forgot password link	Starts reset flow	[Inferred]
E7	Sign in with Google	Social login	[Confirm]
E8	Sign in with SSO	Enterprise SAML or similar	[Confirm]
E9	Start free trial / sign up link	For users without an account	[Confirm]
E10	Error banner / inline errors	Invalid credentials, validation	[Inferred]
E11	Footer links	Terms, privacy policy	[Confirm]
E12	Loading indicator	Seen in app shell	Observed

6. Functional requirements

6.1 Credential login

- FR-1. User enters email and password and selects **Sign in**. On valid credentials, user lands on the dashboard (or the originally requested deep link). [Inferred]
- FR-2. Email field requires a valid email format. Invalid format shows an inline error before submit. [Inferred]
- FR-3. Password field is masked by default. A toggle may reveal it. [Inferred]
- FR-4. Both fields are required. Empty submit shows required-field errors and sends no request. [Inferred]
- FR-5. Invalid credentials show a generic error ("invalid email or password"). The message must not reveal whether the email exists. [Inferred]

- FR-6. Sign in button shows a loading state and disables during the request to prevent double submit. [Inferred]
- FR-7. Pressing Enter in either field submits the form. [Inferred]

6.2 Session

- FR-8. Successful login creates an authenticated session. Logout ends it. [Inferred]
- FR-9. "Remember me" extends session lifetime. Without it the session ends on browser close or at idle timeout. [Confirm]
- FR-10. Opening the login page while already signed in redirects to the dashboard. [Inferred]
- FR-11. Visiting a protected URL while signed out redirects to login, then returns to that URL after login. [Confirm]

6.3 Password recovery

- FR-12. Forgot password opens a reset flow that asks for the account email and sends a reset link. [Inferred]
- FR-13. Response is the same whether or not the email is registered. [Inferred]
- FR-14. Reset link is single-use and expires. [Confirm]

6.4 Alternate login

- FR-15. Google sign-in authenticates via OAuth and maps to an existing VWO account by email. [Confirm]
- FR-16. SSO sign-in asks for the company domain or email and redirects to the identity provider. [Confirm]
- FR-17. Accounts with SSO enforced cannot sign in with a password. [Confirm]

6.5 Two-factor authentication

- FR-18. Accounts with 2FA enabled are asked for a one-time code after correct credentials. Wrong or expired code is rejected. [Confirm]

6.6 Account verification

- FR-19. The app shell contains an "Account Verification Required" banner with a "Verify Now" action. Users with unverified accounts see it after login. [Observed in shell, behaviour Confirm]

7. Security requirements

- SR-1. All traffic over HTTPS. Credentials never in the URL or logs. [Inferred]
- SR-2. Brute-force protection: rate limit, temporary lockout or CAPTCHA after repeated failures. [Confirm]
- SR-3. Inputs sanitised against SQL injection and XSS. Error pages show no stack traces. [Inferred]
- SR-4. Session cookie flagged `Secure` and `HttpOnly`. [Inferred]
- SR-5. Session token rotates on login to prevent fixation. [Inferred]
- SR-6. Password manager autofill must work (correct `autocomplete` attributes). [Inferred]

- SR-7. Open redirect protection on the post-login return URL. [Inferred]

8. Non-functional requirements

Area	Requirement	Tag
Performance	Login page interactive in under 3 s on broadband. Login response under 2 s at normal load.	[Confirm]
Browsers	Latest Chrome, Firefox, Safari, Edge.	[Confirm]
Responsive	Usable on tablet and mobile widths.	[Confirm]
Accessibility	WCAG 2.1 AA: labels, keyboard navigation, focus order, contrast, screen reader error announcement.	[Inferred]
Localisation	English at minimum. Other locales to be confirmed.	[Confirm]
Availability	Matches platform SLA.	[Confirm]

9. User flows

1. **Happy path:** open login, enter email and password, Sign in, dashboard.
2. **Invalid credentials:** enter wrong password, generic error, fields stay filled (except password), retry.
3. **Forgot password:** click link, enter email, receive email, set new password, sign in.
4. **SSO:** click SSO, enter domain, identity provider, return to VWO, dashboard.
5. **2FA:** correct credentials, enter code, dashboard.

10. Error and edge states

Case	Expected
Empty fields	Inline required errors
Bad email format	Inline format error
Wrong password	Generic credentials error
Unknown email	Same generic error
Account locked	Clear lockout message with next step
Unverified account	Verification banner after login
Network failure	Retryable error, no data loss
Server 5xx	Friendly error, no stack trace
Session expired	Redirect to login with message

11. Analytics (suggested)

Track: login attempts, success rate, failure reasons, method used (password, Google, SSO), forgot-password starts, time to login. [Confirm]

12. Acceptance criteria (summary)

- AC-1. Valid user signs in and reaches the dashboard.
- AC-2. Invalid input never signs a user in and never exposes whether an email exists.
- AC-3. Password reset works end to end and the link cannot be reused.
- AC-4. Protected pages are inaccessible without a session.
- AC-5. Page passes the accessibility and browser checks in section 8.

13. Open questions

1. Which login methods are live: Google, SSO, others?
2. Is "Remember me" present, and what are the session lifetimes?
3. What is the lockout or CAPTCHA threshold?
4. Is 2FA mandatory, optional or per-account?
5. Which locales and browsers are in scope?
6. What are the exact error message texts?
7. What does the "Verify Now" flow require?

14. Next steps

1. Load the page in a real browser, capture the actual elements, and update section 5.
2. Resolve section 13 with the product team.
3. Derive test cases from sections 6 to 10.